

Common Problems (and Solutions) for Developing Saxophonists

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Abstract: In this article, I address three major concerns music teachers often present to me when I work with their saxophone students: tone, intonation, and articulation. After addressing issues concerning equipment, I discuss the likely causes of various problems within these broad categories, then offer practical solutions for the classroom, including many tips and exercises.

The Young Saxophonists' Plight - Tone, Tuning, and Articulation

When I present clinics for middle and high school saxophone students, I am often greeted by music teachers who lament the state of their saxophone section. While young saxophonists are often among the first to make a sound in a beginning band class, they can lag behind the other instruments in playing quality if their unique challenges are not fully understood. After many years of discussing saxophone problems with music teachers in Canada and the United States, I have seen common threads. In this article, I will discuss three of the most common issues I hear from music teachers about their saxophone section, some possible causes, and how I approach solving those problems.

A Note on Equipment

I know that music teachers are often constrained by very limited budgets, but I always give the same piece of advice when it comes to improving a saxophone section: your students need professional model mouthpieces. If you have the funds to buy one new saxophone or ten new mouthpieces, buy ten new mouthpieces. I am often amazed by the prevalence of stock mouthpieces in band classrooms; the difference between one of these mouthpieces and a professional quality mouthpiece is enormous.

When working with my own students, I tell parents that the mouthpiece is the instrument, and the saxophone is just a sophisticated amplifier. While it is possible to make a student model saxophone sound lovely with a professional model mouthpiece, I would not be able to do the same with a stock mouthpiece on my own professional model instrument.

In addition to good quality mouthpieces, students need high-quality, undamaged reeds to play successfully. Reeds that are warped, chipped, too hard, too soft, moldy, or otherwise damaged will not make a good sound. I encourage my students to have at least three working reeds at all times, rotating them for each practice session to extend their lives. They must be



properly stored (either in the case they come in or a reed case – not in a plastic bag or loose in the case!), removed from the mouthpiece after every playing session, and replaced when they are damaged or become too soft from use.

Though the instrument itself is not the top priority for playing well, if the student has an instrument that is in poor repair, it will be impossible for them to be successful. Leaks, missing screws, dented bells, missing felts, bent keys, broken or unhooked springs, and damaged corks are all too common, unfortunately. Having students swab instruments after each use, consistently using the neck cap, regularly applying cork grease, and visually checking the instrument for anything that might look unusual are steps students can be expected to do themselves; teachers must also regularly check for loose screws, make sure springs are properly placed, and oil the keys at least once a year. Instrument maintenance is beyond the scope of this article, but there are many resources available on caring for saxophones available elsewhere.

Making sure students have high quality, well-maintained equipment is essential for troubleshooting – if they have poor equipment, no amount of good pedagogy will solve their problems!

Problem 1: Honky, thin, abrasive, or otherwise unpleasant tone that can't blend in with the rest of the band

I often tell my students that the saxophone is the easiest band instrument to play, but one of the hardest to master. Owing to the well-designed key work and relatively simple sound production mechanism, the vast majority of students have success making sound on a saxophone on the first day of playing. The problem, then, is the quality of that sound, which often leaves much to be desired at first. While honky, thin, and abrasive sounds are caused by slightly different playing considerations, most can be attributed to a combination of embouchure and air issues.

Biting

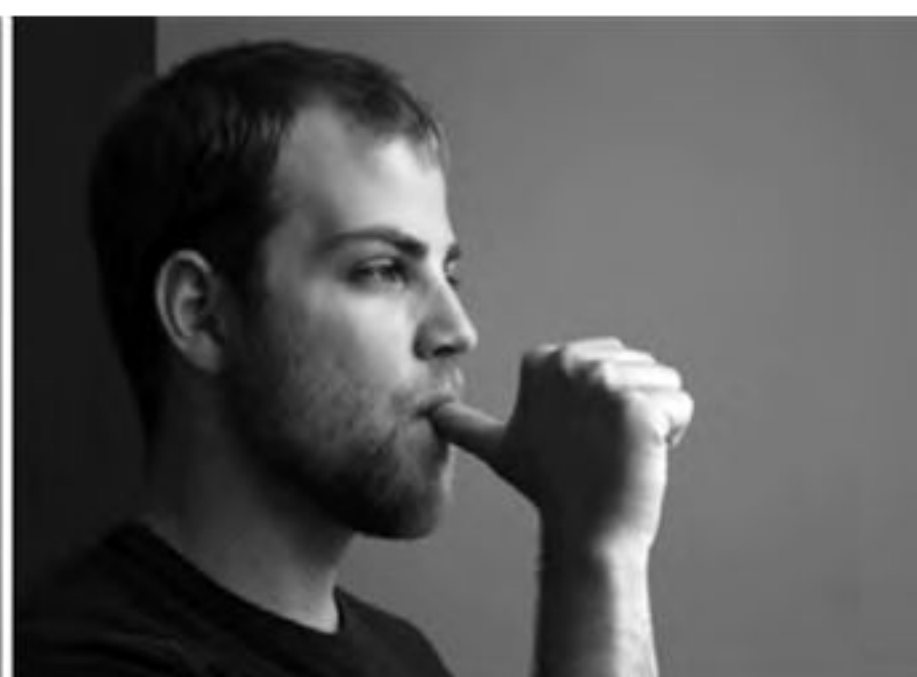
Biting is one of the most common problems for young saxophonists. Biting involves using too much pressure from the bottom jaw to support the embouchure, resulting in a pinched sound and sharp pitch overall. I find the easiest way to deal with biting is to prevent it in the first place, but re-explaining proper embouchure formation can resolve biting problems gradually.



Step 1.



Step 2.



Step 3.



Step 4. Repeat steps 1-3 with the saxophone neck and mouthpiece



Exercise (Embouchure Sequence):

I use the following sequence for teaching basic embouchure to my students:

- 1 Starting with the thumb as a stand-in mouthpiece, have students put their top teeth on the pad of the thumb, feeling the weight of their head pressing downward (this helps prevent pushing upwards with the bottom jaw).
- 2 With the top teeth resting on the thumb, have the students say “ooo” (this brings the corners of the mouth forward, drops the bottom jaw, and helps to engage the muscles of the lower lip for the next step).
- 3 Have students roll about half of their bottom lip over their teeth, thinking of the lip as a cushion or pillow for the reed (note that to create a cushion with the lip, the muscles will have to be somewhat bunched and thus firm).
- 4 Repeat the same procedure, using the neck and mouthpiece instead of the thumb. Make sure to show students how much mouthpiece to take in (see below for a more thorough explanation).

I find that the majority of embouchure problems can be prevented if this explanation is used and reinforced from day one. If the students have a relatively well-formed embouchure, they will still need occasional reminders. If students are used to thinking of the bottom lip as a soft and flexible support for the reed, reminding them to “keep the cushion” is often helpful.

It is important to note that students bite because it is physically easier to produce a sound when biting than with proper embouchure support. Because the jaw muscles are much stronger than the cheek muscles, the cheeks fatigue much faster, and students will resort to the jaw to continue playing. If this becomes a habit, biting becomes a regular part of the embouchure, and becomes more difficult to correct as this problem becomes engrained. Therefore, students must be carefully monitored when playing, and continually reminded to engage the muscles in their cheeks, not just their jaws. I often say, “push your corners toward the mouthpiece,” “remember to say ‘ooo,’” “spread your teeth apart,” “stretch an imaginary elastic attached to your

nose and chin” or “keep the cushion” in the bottom lip when I see a student whom I believe is biting.

Amount of Mouthpiece

The amount of mouthpiece students take in is an important consideration. When a student is playing with a honky sound, they are likely playing with too much mouthpiece in their mouth; when the sound is thin, they are often using too little.

Exercise (The Paper Trick):

To find out how much mouthpiece the student needs to take in, take a piece of paper and slide it between the reed and the mouthpiece. The piece of paper will naturally fall where the reed and mouthpiece come in contact with each other (there is no need to force the paper down farther than it goes with no pressure). Since one of the primary goals of the embouchure is to create a seal, finding this point tells the student how much mouthpiece to take in. You can then take a rubber mouthpiece patch, cut it in four pieces, and place one of these sections at the spot where the reed and mouthpiece meet. This small piece of rubber gives the student a consistent spot to place their teeth, and prevents reverting back to old habits. By making the strip of rubber thin, students will immediately know if they’ve taken in too much or too little mouthpiece, because they will not be on the patch. If the mouthpieces are on hand before classes begin, teachers could put these patches on in advance, which stops this issue before it starts.

While giving the students a consistent point of reference for the amount of mouthpiece to take in is helpful, students must also be exposed to characteristic sound examples and asked to listen to their own sounds as they play. Thus, the amount of mouthpiece they take in can serve as an opportunity to listen for their best sound.

Exercise (Experiment and Listen!):

Since the amount of mouthpiece taken in has a dramatic impact on sound quality, students can take in too much mouthpiece, too little mouthpiece, and find the ‘sweet spot’ by listening to the difference when they play in each position. Have students



Paper between the reed and mouthpiece



Do not push the paper down



Approximate size and placement of rubber patch

experiment with all three options, and ask them to describe the sound for each. Then, tell them to always be listening to themselves while playing, making sure they are always playing at the ‘sweet spot’ for the best sound quality.

Mouthpiece Pitch

Mouthpiece pitch is also an important indicator of biting. Each instrument in the saxophone family has a ‘proper’ mouthpiece pitch, and significant variation from this pitch in beginners often indicates a larger problem. Having students play on the mouthpiece alone is noisy, but it is very helpful for proper embouchure development. In addition to diagnosing embouchure problems, an inability to produce a consistent mouthpiece pitch can indicate inadequate air pressure.

As a general rule, students will want to aim for the following pitches on their mouthpiece alone at a fortissimo dynamic:



Provided that they’re playing a good quality mouthpiece and reed, students who are biting will hit pitches higher than those specified above. Often, I hear students being instructed, “not to bite,” but I often wonder if they know precisely what this means. It is quite difficult to stop biting if it is a habit, because biting affects more than just the amount of pressure being exerted on the mouthpiece. In particular, a decrease in embouchure pressure means they will need more air pressure to make a characteristic sound, which is not immediately obvious to students (the air exercises described below will help students increase their air pressure).

In terms of diagnosing biting problems, there are some characteristics of the saxophone that can help. Because the saxophone has an octave key, the ability to move quickly between octaves without difficulty is an important indicator of correct embouchure and air. If a student is playing with the proper amount of embouchure pressure and air pressure, they should be able to play in all octaves of the instrument without making changes to their embouchure. I sometimes hear teachers tell students to “drop the jaw” to play lower in the range or “lip up” in the high range, but this is not the best method for playing saxophone with tonal consistency. The jaw should be in a relatively neutral position, neither pushed forward nor pulled back, where students can play in all registers without movement.

Exercise (Check the Octaves):

The ability to jump octaves quickly is a great check for biting using the range of the instrument. I find the best way to do this is to flick the octave key for students: have them finger a low G,

playing at a mezzo forte dynamic, then lightly ‘tap’ the octave key for them several times while they are playing.



Have the student play a whole note, then flick the octave key. The pitches will jump up and quickly back down if the student is playing properly.

What we are listening for is a relatively quick jump back down to the low G without articulation (it may ‘hang on’ in the upper octave for a second or two, but it should fall relatively quickly). If the pitch stays in the top octave, there are essentially two options:

The student is biting consistently.

The student is trying to change their embouchure when changing octaves (likely firming up for the top note and relaxing in the bottom).

Flicking the octave key without telling the student when you will do it, or what the goal of the exercise is, can often prevent option #2. In addition, make sure the student understands that the embouchure should not change significantly register to register. This exercise can help show students learn how little they should have to do to change octaves.

If a student is biting consistently, it won’t matter if they know you are flicking the octave key for them – the pitch will either never come back down or stay up for an extended period of time. If this is the case, make sure to re-explain the embouchure, and in particular the need for cushion from the lower lip.

Air

It is all but impossible to have a discussion of good tone without discussing air. Often, students will not be able to form a good embouchure without having a supported air column, and thus good tone is impossible. Particularly if the student has been biting, it is essential to make sure they are using lots of well-supported air to make up for the pressure difference. In my experience, the saxophone sounds best when played with fast, cold air that is high in the mouth.

The term ‘air support’ can be unclear to students in itself, so I often also say the air is focused and pressurized. It is focused if the student is filling the saxophone with air (versus filling the mouth with air); cheek puffing is a sure sign of inefficient air use. Students can also think about directing their air into the mouthpiece. When the air is supported, the core muscles are engaged while playing – if the core is not engaged, the air will not have enough pressure behind it to produce a characteristic sound.

Exercise (Fox):

To get saxophonists using an efficient airstream, I have them say “fox,” emphasizing the “f” syllable (“Fffffffox”). This creates the kind of air pressure needed to play saxophone, puts the tongue in a relatively neutral position, and gives students an easy to remember physical sensation. I also say “put your air behind your two front teeth,” as this is essentially what happens when we use the “fox” syllable.

Problem 2: Intonation that’s all over the map — but especially sharp!

In my experience, the vast majority of tuning problems are caused by embouchure problems. If the student is playing with a good embouchure and air support, most intonation problems are solved on their own. There are a few things to look for, however, which indicate an overall problem.

Visual cues

The neck cork can give an important visual checkpoint for knowing what sort of problems students are having. As a general rule, about one fourth of the cork should be showing on the neck when the mouthpiece is attached (this can vary slightly). If your students are playing so far out on the cork that the mouthpiece is in danger of falling off in order to play in tune, you can be virtually certain they are biting. All of the embouchure and air exercises above should help the student develop a more in tune (and in tone) sound, provided that they are using good-quality equipment.

Conversely, if the mouthpiece is pushed in so far that there is no cork showing, the student is likely playing with a wide, ‘honky’ air column, causing a low pitch center. If the student is in general playing flat, they are likely not using enough air support/pressure, and could also be using too much mouthpiece.

Again, note that a poor quality mouthpiece and reed will affect intonation, so students with poor equipment, damaged reeds, and the incorrect reed strength will have more intonation issues.

It is important to remember that even if the student is playing with a good embouchure and a reasonable amount of cork showing, the instrument will not be completely in tune because no saxophone is completely in tune with itself. In general, the saxophone has a tendency to be sharp in the lowest and highest registers and flat in the notes near the bottom of the staff.

There are some corrective fingerings available for problematic pitches, but they should be viewed as a last resort. Generally, students should be expected to listen for pitch and develop over time. In particular, students must be encouraged to actively listen to their own playing and to other, in-tune sources. In addition, the saxophone is best tuned to written F-sharp with the octave key on all instruments (Concert A for alto/baritone, Concert E for soprano/tenor), as this is the most telling pitch for overall intonation. Simply teaching saxophonists to quickly tune to their written F-sharp against a piano, before tuning with everyone else, can make a major difference.

Problem 3: Aggressive, unclear, slow, or otherwise unpleasant articulation

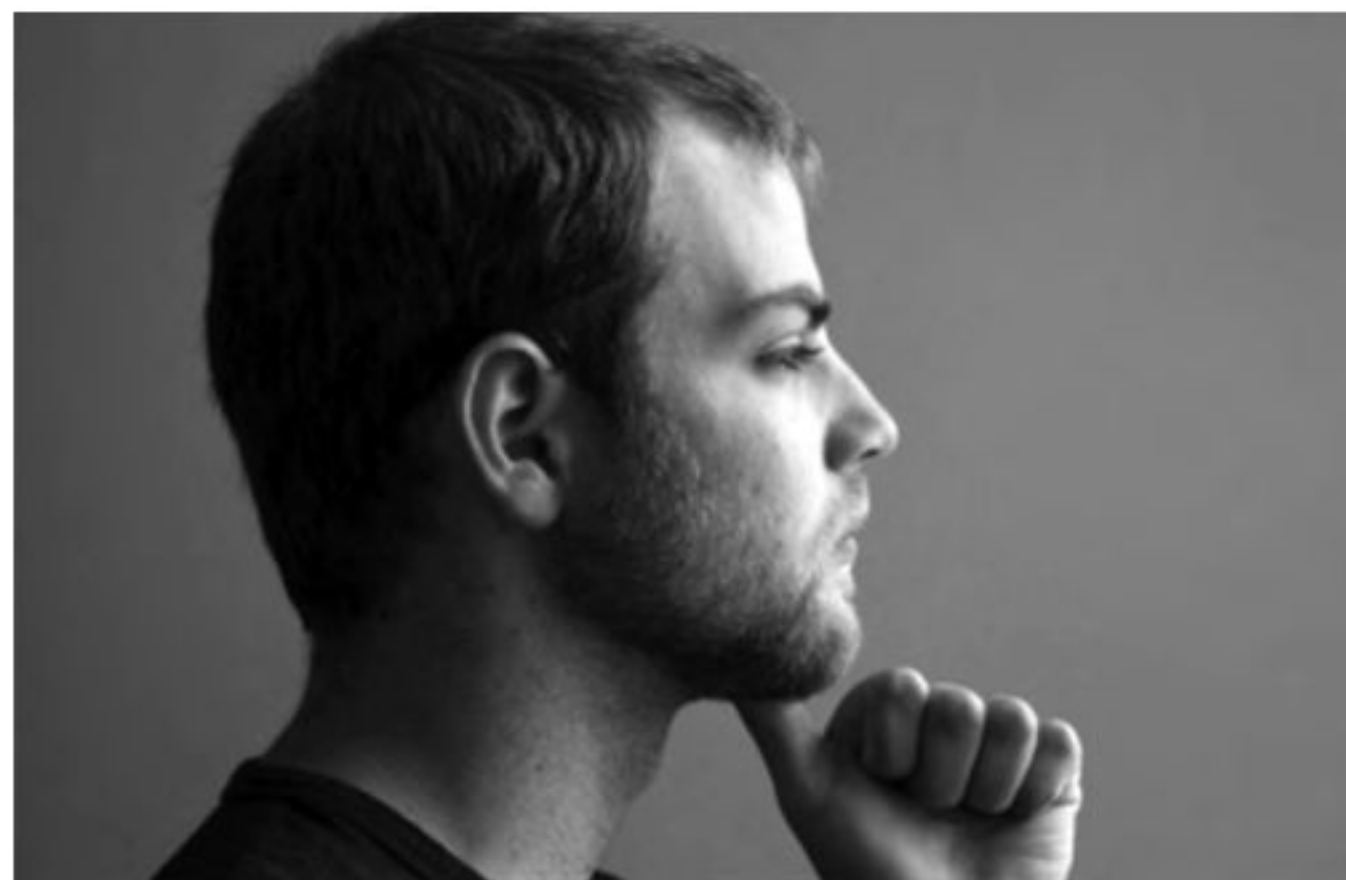
I frequently hear three articulation problems that involve not touching the reed in the first place, which I will colloquially call “hoo,” “woo,” and “koo,” articulation. In addition, I often hear aggressive, “thuddy” or “spitty” articulation, which involves touching the reed with the wrong part of the tongue, trying to fill the space between the reed/mouthpiece with the tongue, or tonguing forcefully. I will address the common causes of these problems, and how I work with students to get back on the right track.

A “hoo” articulation involves stopping the air between pitches. This is both common and relatively easy to diagnose in beginners, as they will have audible space between pitches. A “woo” articulation involves moving the jaw or ‘chewing’ to create separation between pitches. A “koo” articulation involves using a glottal stop at the back of the throat, rather than actually touching the reed with the tongue. I find this can be the most difficult to diagnose, as students are often incredible at copying a true articulation sound. When students are saying “koo” instead of touching the reed, you will often hear a grunt or vocal sound when they first start playing.

Though most band directors teach articulation with the syllable “ta” or “da,” I suggest using “la” as the articulation syllable for saxophone. The biggest problem I see with these syllables is that they encourage the student to move too much of the tongue, and to think of the articulation as explosive or aggressive.

Exercise (Check the Jaw):

Have the students put their thumbs under their chins, behind the jawbone (where there is just soft tissue).



Place the thumb in the soft tissue right behind the jaw bone

Have them say “ta” and “da” several times in a row, noting how much movement they feel. Then, have students say “la” several times, noting there is little to no movement—meaning there is very little movement in the back of the tongue.

The one drawback to using a “la” syllable for saxophone is that it can encourage sluggish articulation with too much reed



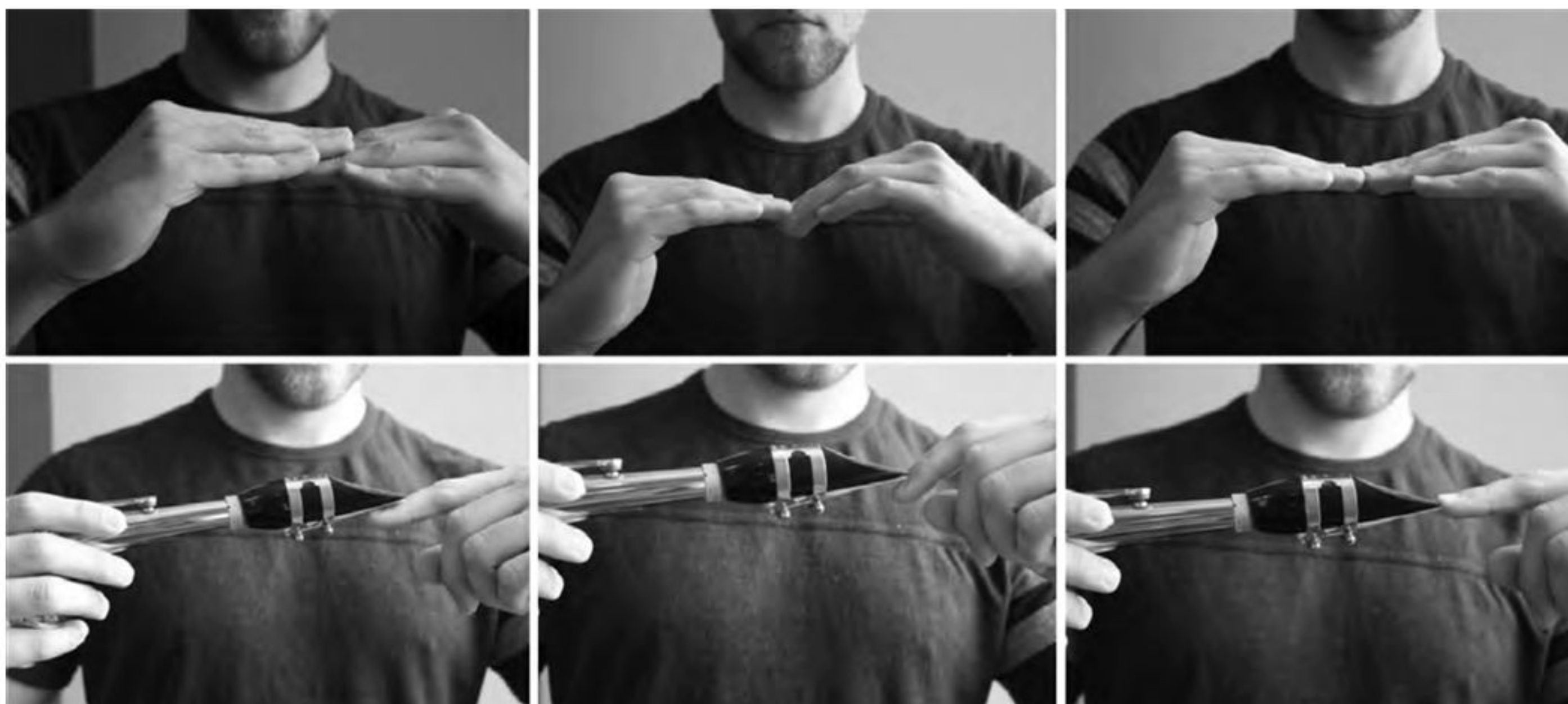
1. Too much cork



2. Too little cork



3. Approximately the right amount



1. Touching too much of the reed between the mouthpiece (not recommended) 2. Recommended placement

3. Tip to tip or filling the space

contact if not explained well. To prevent this, I frequently use my hands as a model for tongue placement. If you imagine one hand is the tongue and the other is the reed, you can provide a visual model for what is happening internally. Alternatively, you can use the saxophone and your hand if this demonstrates more clearly.

It is easy to see when students are moving too much of their tongues: if you see any visible movement under the chin during articulation, the student is moving their tongue too much. In particular, students need to know that they can move the front of the tongue without having movement in the back.

Exercise (Taste Test):

Brand new reeds have a particular woody taste, which you can use to your advantage when students are not touching the reed. I often tell students to “touch the tip of the reed with the tip of your tongue that has taste buds,” which means they can literally taste their articulation if they are touching the reed. I talk specifically about the taste buds because some students will either try to use the absolute tip of the tongue, or will use the underside of the tongue.

Articulation checks in groups

Because we can't always spend the time to assess every student individually in the band classroom, it is important to have methods for assessment that allow individuals to play, while engaging the rest of the class.

Exercise (Echo Games):

To assess articulation without putting undue pressure on individual students, I like to play echo games. For example, I tell the students to play a 4-beat rhythmic pattern of their choosing on do, mi, or sol in whatever key we're playing. Each individual student plays a pattern, followed by the entire band or section copying that pattern. This requires the rest of the band to develop some basic aural skills (they need to hear which pitch was played and play the rhythmic pattern), but it also gives the teacher an important opportunity to assess the articulation of individual students. Rather than ‘go down the line’ and listen to articulations in a stressful situation, students get the opportunity to think creatively while the teacher can diagnose problems in a lower stress setting.

Final Thoughts

Although one article cannot hope to solve all of music teacher's saxophone woes, this should provide some valuable information for solving problems. There are many other factors to consider in the art of teaching saxophone, but those mentioned here are so prevalent that I feel they constitute a good starting place. With quality equipment and saxophone-specific teaching strategies, all students can be successful saxophone players.



Jennifer Blackwell is a Canadian saxophonist and music educator. As an advocate for new music, she has premiered numerous new works for saxophone at many venues, including the North American Saxophone Alliance's (NASA) Biennial Conference in 2014. She serves as Editor of NASA's scholarly publication, *The Saxophone Symposium*, and is co-chair of Gender Research in Music Education. Jennifer holds a bachelors degree in music education from the University of Western Ontario, and masters degrees in saxophone performance and music education from Central Michigan University. She is currently pursuing her PhD in Music Education with a minor in Saxophone Performance at the Indiana University Jacobs School of Music, where she teaches woodwind methods and rehearsal techniques. For more information, please visit jblackwellmusic.com.

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